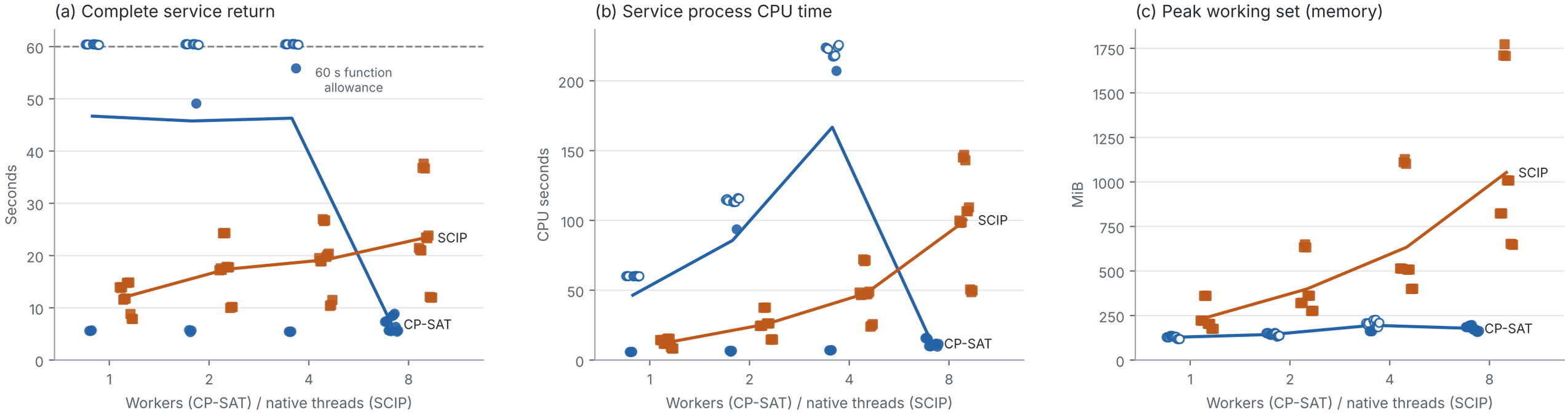
Worker/thread sweep | how each solver responds to added workers

4 event–setting pairs × 3 executions = 12 runs per solver and level; 60-second allowance; lines join level means



Hollow markers are complete candidates returned 25–126 ms after the 60-second boundary without a proof; their times and resource use are retained. CP-SAT with 1–4 workers searched the full allowance; with 8 workers it proved every optimum quickly. SCIP proved every optimum at every level but became slower and used more CPU and memory as threads were added. CPU time and peak working set are whole-process measurements from fresh workers, including native solver threads.